



NORTH SYDNEY GIRLS HIGH SCHOOL

YEAR 12 – TERM 1 ASSESSMENT

MATHEMATICS EXTENSION 2

2008

TIME ALLOWED: 60 minutes
Plus 2 minutes reading time

NAME _____

CLASS _____

Q 1 / 16	Q 2 / 14	Q 3 / 11	Total / 41

INSTRUCTIONS:

- Start each question on a **NEW PAGE**
- Hand in each question separately, including a sheet for non-attempts
- Show all necessary working
- Write on one side of the page only
- Marks may be deducted for incomplete or poorly arranged work

Question 1 (16 marks)

a) Let $z = 2 + i$ and $w = 1 - 2i$. Find, in the form $x + iy$,

(i) $z + w$ **1**

(ii) $\frac{z}{w}$ **2**

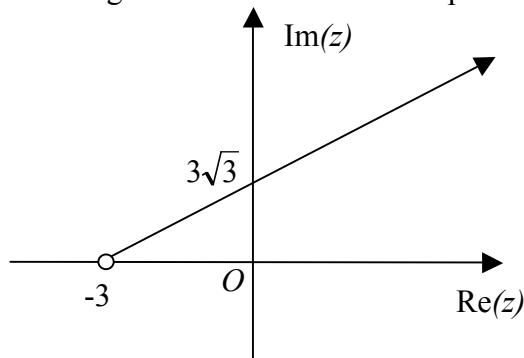
(iii) $\text{Im}(z + \bar{w})$ **1**

b) (i) For the complex number $z = \sqrt{3} + i$ find $|z|$ and $\arg z$.
Hence express z in the form $r(\cos \theta + i \sin \theta)$. **2**

(ii) Find $\bar{z}$ and z^2 giving your answers in the form $a + ib$. **2**

(iii) Use De Moivre's Theorem to find a positive integer m such that $(\sqrt{3} + i)^m - (\sqrt{3} - i)^m = 0$. **3**

c) The diagram shows the locus of a point P representing the complex number z .



(i) Find the equation of the locus of P in terms of z . **1**

(ii) Find the modulus and argument of z when $|z|$ takes its least value. **2**

(iii) Hence find z in the form $a + ib$ for which $|z|$ is least. **2**

Question 2 (14 marks)

- a)** (i) On the same diagram, draw a neat sketch of the locus specified by each of the following: 2

(α) $|z + 1 - 3i| = 2$

(β) $|z - 5i| = |z + i|$ 2

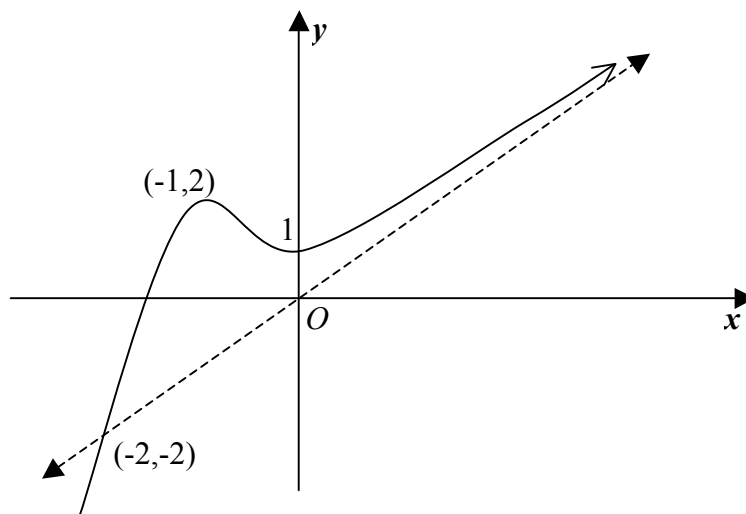
- (ii) Hence shade the region of points z which satisfies simultaneously 1

$$|z + 1 - 3i| \leq 2 \quad \text{and} \quad |z - 5i| \geq |z + i|$$

- b)** On the same number plane, sketch the graph of $y = \frac{1}{x-3}$ and hence the graph of 3

$y = \frac{1}{\sqrt{x-3}}$, clearly indicating any asymptotes and intercepts.

- c)** The diagram shows the graph of $y = f(x)$. The line $y = x$ is an asymptote.



Draw neat sketches of the following *on the answer sheet provided*, showing clearly any asymptotes and important features. 2

(i) $f(|x|)$ 2

(ii) $\frac{1}{f(x)}$ 2

(iii) $xf(x)$ 2

Question 3 (11 marks)

- a)** If the domain of the function f given by $f(x) = \frac{1}{1-x^2}$ is restricted to $\{x : |x| > 1\}$, which one of the following describes the range of f ? Briefly justify your answer. 2

(A) $-\infty < f(x) < -1$

(B) $-\infty < f(x) < 0$

(C) $-\infty < f(x) < 1$

(D) $-1 < f(x) < \infty$

(E) $0 < f(x) < \infty$

- b)** The graph of $y^2 = x^2 + 9$

I has the x axis as an axis of symmetry

II has the y axis as an axis of symmetry

III has point symmetry about the origin

Which of the above statements is true? Justify your answer with a sketch. 2

- (A) I only (B) II only (C) III only (D) I and II only (E) I, II and III

- c)** Sketch the roots of $x^6 - 1 = 0$ on the Argand diagram. 2

- d)** The points P and Q represent the complex numbers $4 - 2i$ and $6 + 4i$ respectively.

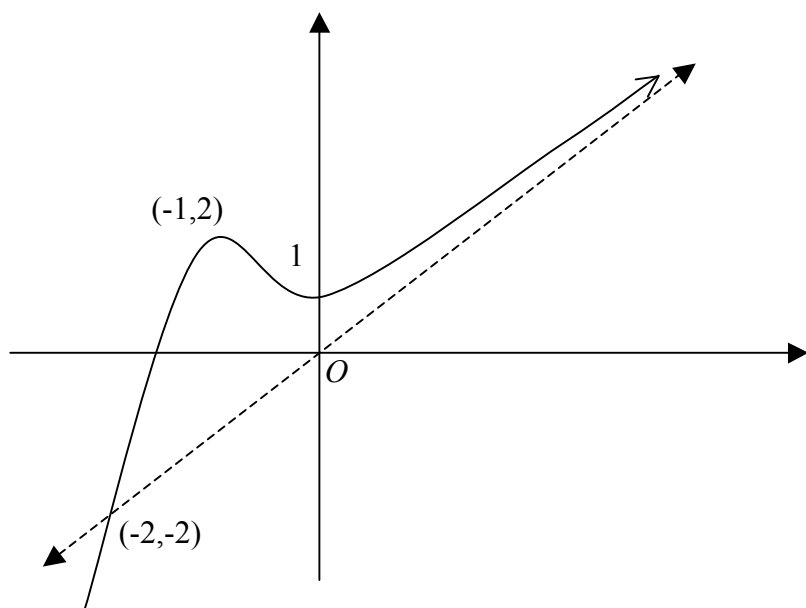
(i) Find the complex number represented by vector $\overrightarrow{PQ}$ 1

(ii) The points P , Q and S form an equilateral triangle.
Find a possible value of the complex number represented by S . 3

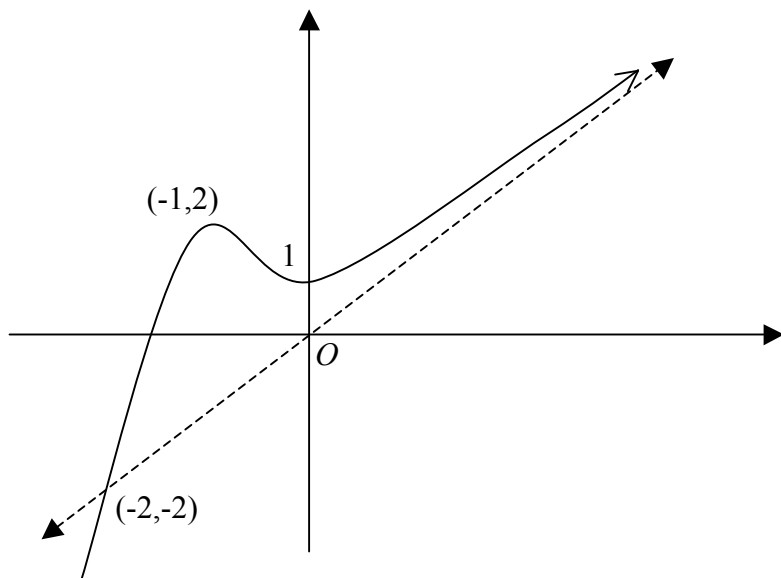
(iii) The points P , O , Q and R (named in cyclic order, O being the origin) form a parallelogram. Find the complex number represented by R . 1

End of paper

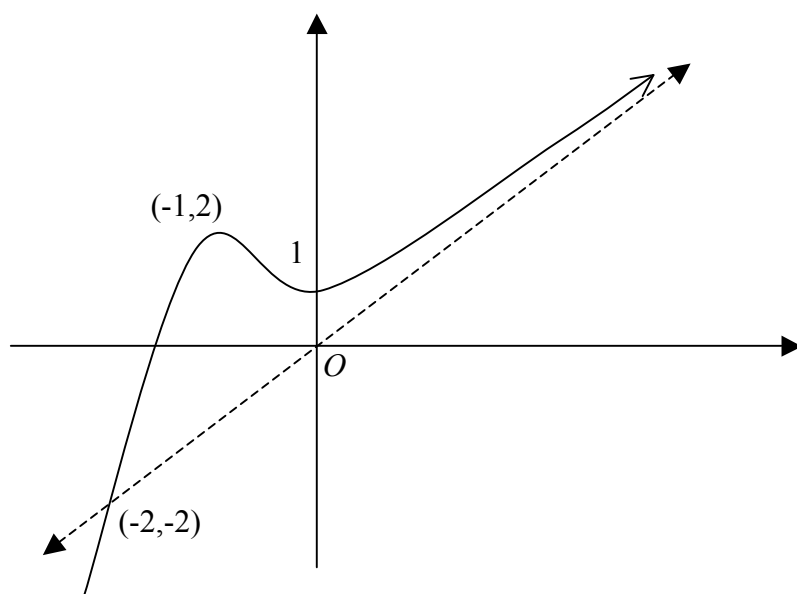
(i)



(ii)



(iii)



Solutions

Question 1

- (a) (i) $z + w = 2 + i + 1 - 2i = 3 - i$
 (ii) $\frac{z}{w} = \frac{2+i}{1-2i} \times \frac{1+2i}{1+2i} = \frac{2+i+4i-2}{5} = i$
 (iii) $\text{Im}(z + \bar{w}) = \text{Im}(2 + i + 1 + 2i) = 3$

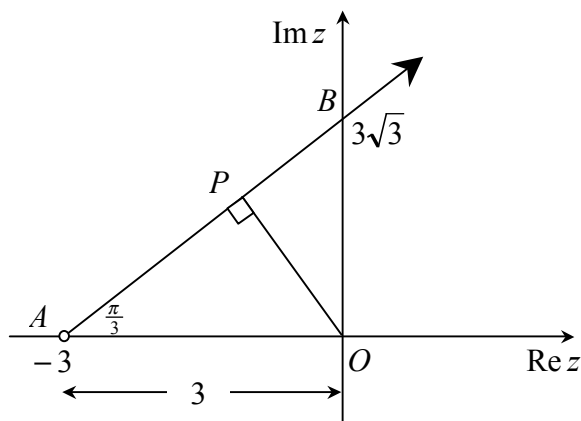
- (b) (i) $z = \sqrt{3} + i$
 $|z| = 2, \arg z = \frac{\pi}{6}$
 $\therefore z = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$

- (ii) $\bar{z} = \sqrt{3} - i$
 $z^2 = \left[2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right]^2$
 $= 4 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$ [By de Moivre's Thm]
 $= 4 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$
 $= 2 + 2i\sqrt{3}$

- (iii) $(\sqrt{3} + i)^m - (\sqrt{3} - i)^m = 0$
 $\therefore \left[2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right]^m - \left[2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right) \right]^m = 0$
 $\therefore 2^m \left[\cos \frac{m\pi}{6} + i \sin \frac{m\pi}{6} - \cos \left(-\frac{m\pi}{6} \right) - i \sin \left(-\frac{m\pi}{6} \right) \right] = 0$
 $\therefore \cos \frac{m\pi}{6} - \cos \left(\frac{m\pi}{6} \right) + 2i \sin \frac{m\pi}{6} = 0$
 $\therefore \sin \frac{m\pi}{6} = 0$
 $\therefore \frac{m\pi}{6} = n\pi + (-1)^n \times \sin^{-1}(0)$
 $\therefore m = 6n$
 ie m is a multiple of 6

- (c) (i) $\arg(z + 3) = \theta$ where $\tan \theta = \sqrt{3}$
 $\therefore \arg(z + 3) = \frac{\pi}{3}$

(ii)



Let P represent z where $OP = \min|z|$ and so $OP \perp AB$

$$\therefore \sin \frac{\pi}{3} = \frac{OP}{3}$$

$$\therefore OP = 3 \sin \frac{\pi}{3} = \frac{3\sqrt{3}}{2}$$

$$\therefore \min|z| = \frac{3\sqrt{3}}{2}$$

$$\angle POX = \frac{5\pi}{6} \Rightarrow \arg z = \frac{5\pi}{6}$$

$$\begin{aligned} \text{(iii)} \quad z_{\min|z|} &= \frac{3\sqrt{3}}{2} \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \\ &= \frac{3\sqrt{3}}{2} \left(-\frac{\sqrt{3}}{2} + \frac{i}{2} \right) \\ &= -\frac{9}{4} + \frac{3i\sqrt{3}}{4} \end{aligned}$$

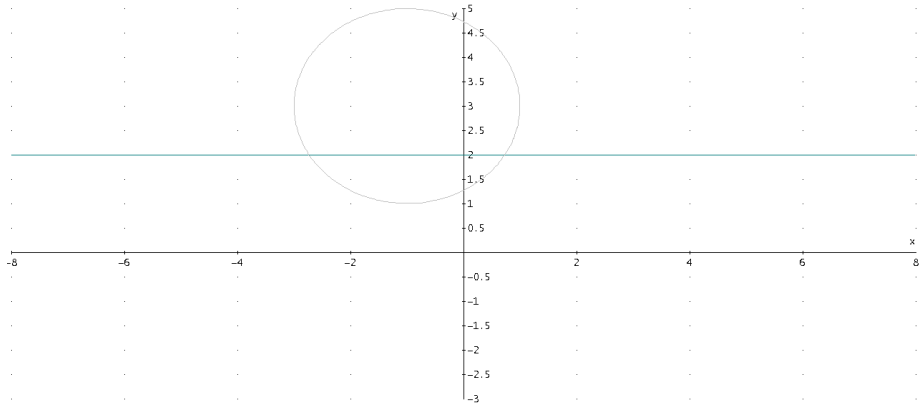
Question 2

(a) (i) $|z+1-3i| = 2 \Rightarrow |z - (-1+3i)| = 2$

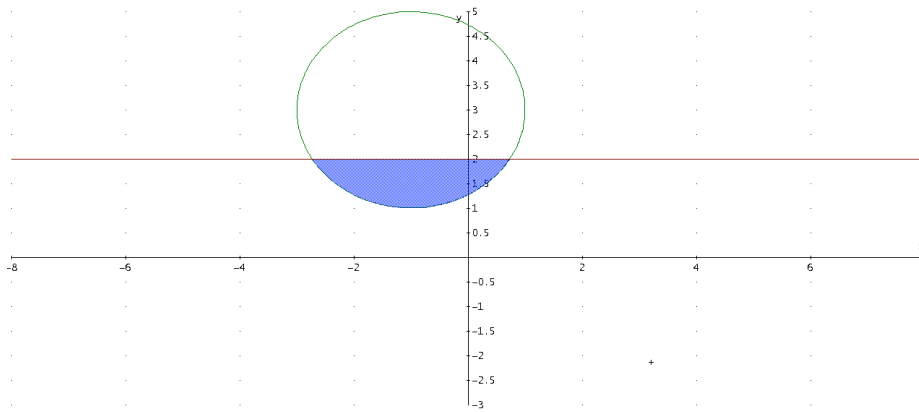
This is a circle of radius 2 centred at $z = -1+3i$

$$|z-5i| = |z+i|$$

This is the perpendicular bisector of $z = 5i$ and $z = -i$ ie $z = 2$

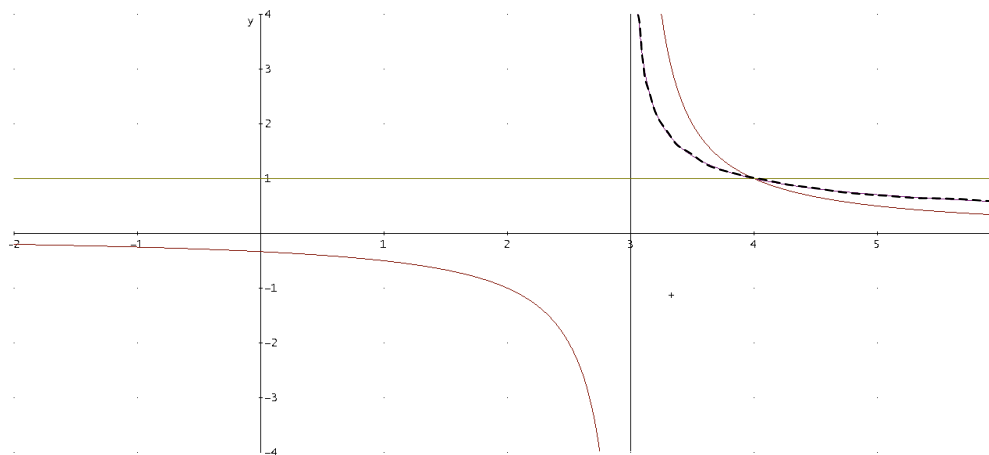


(ii)



(b) The curve drawn with dashes is $y = \frac{1}{\sqrt{x-3}}$

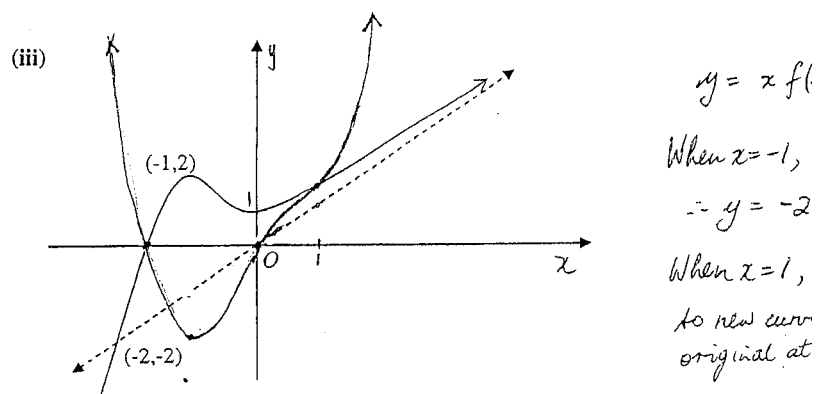
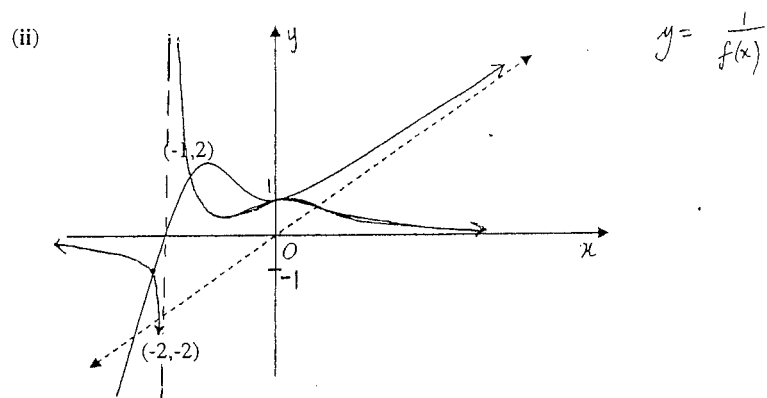
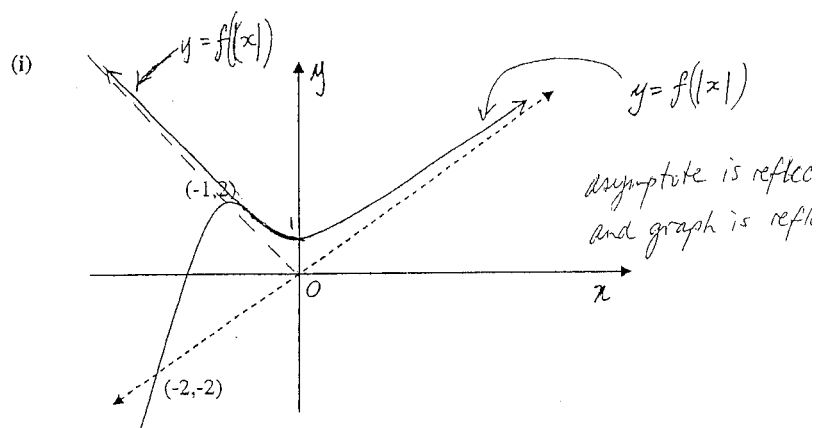
The x – axis is a horizontal asymptote



(c)

Answer Sheet for Question 2 c)

Name: _____

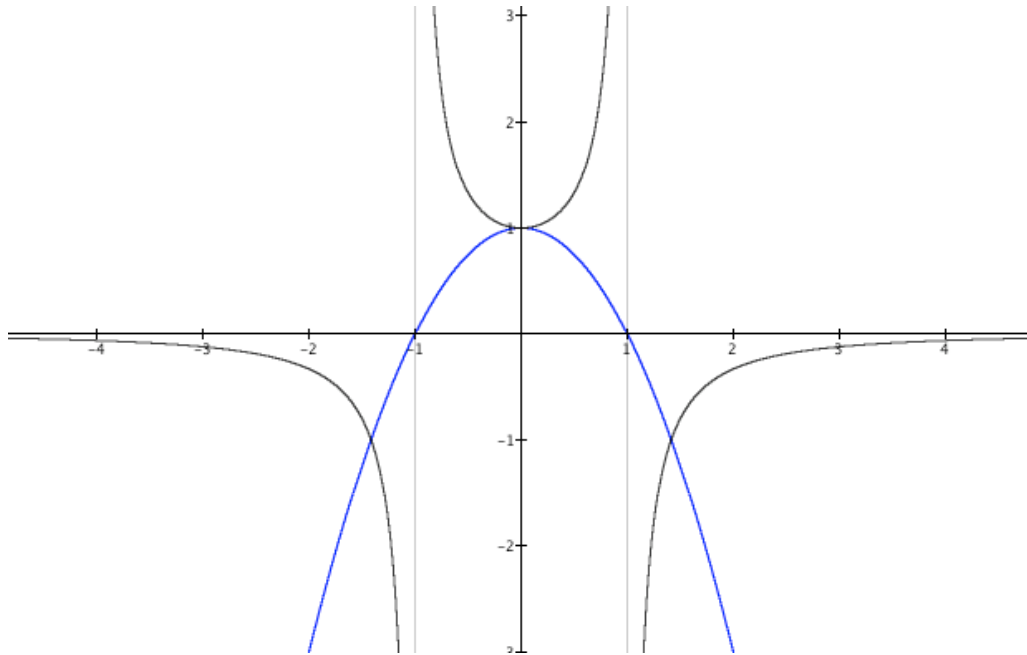


Question 3

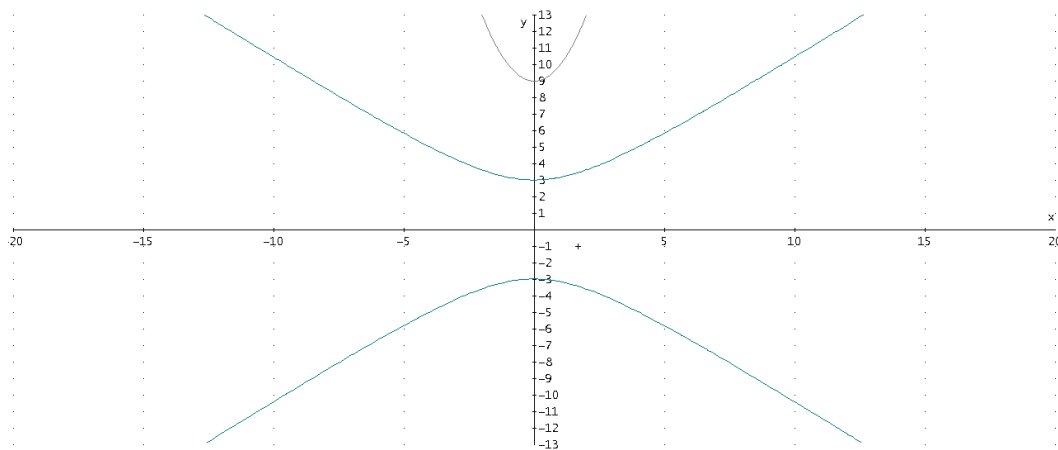
(a) $f(x) = \frac{1}{1-x^2}$

For $|x| > 1$, then $-\infty < f(x) < 0$

$\therefore$ B



(b)



From the graph $y^2 = x^2 + 9$ is symmetric in the y and x axes, and is also point symmetric about the origin. So I, II and III ie **E**

(c) $x^6 - 1 = 0$

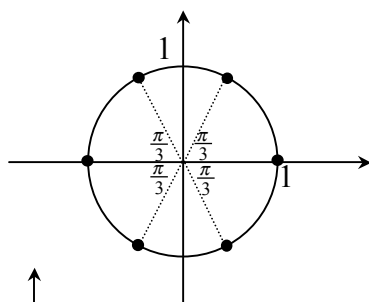
$$\therefore x^6 = 1 = 1\text{cis}(0 + 2k\pi) = \text{cis}(2k\pi), \quad k \in \mathbb{Z}$$

$$\therefore x = \text{cis}\left(\frac{2k\pi}{6}\right) = \text{cis}\left(\frac{k\pi}{3}\right), \quad 0 \leq k \leq 5$$

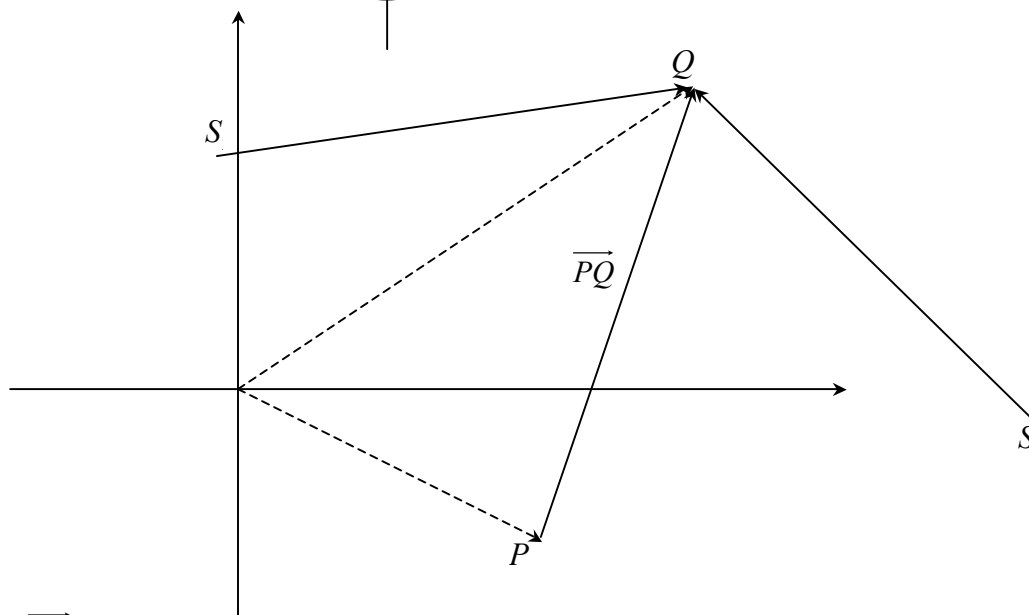
$$\therefore x = 1, \text{cis}\left(\frac{\pi}{3}\right), \text{cis}\left(\frac{2\pi}{3}\right), \text{cis}(\pi), \text{cis}\left(\frac{4\pi}{3}\right) = \text{cis}\left(-\frac{\pi}{3}\right), \text{cis}\left(\frac{5\pi}{3}\right) = \text{cis}\left(-\frac{2\pi}{3}\right)$$

So the roots all lie on the unit circle and are equally spaced around it, by 60° .

The non-real roots occur in conjugate pairs



(d)



(i) $\overrightarrow{PQ}$ is represented by $6 + 4i - (4 - 2i) = 2 + 6i$

(ii) Two points can be found S_1 or S_2 , by rotating $\overrightarrow{PQ}$ by 60° clockwise or anti-clockwise.

This gets the vectors $\overrightarrow{S_1Q}$ and $\overrightarrow{S_2Q}$.

$$\begin{aligned} \overrightarrow{S_1Q}: \quad (2 + 6i)\text{cis}60^\circ &= (2 + 6i)\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) \\ &= (1 + 3i)(1 + i\sqrt{3}) = 1 - 3\sqrt{3} + i(3 + \sqrt{3}) \end{aligned}$$

$$\begin{aligned} \overrightarrow{S_2Q}: \quad (2 + 6i)\text{cis}(-60^\circ) &= (2 + 6i)\left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right) \\ &= (1 + 3i)(1 - i\sqrt{3}) = 1 + 3\sqrt{3} + i(3 - \sqrt{3}) \end{aligned}$$

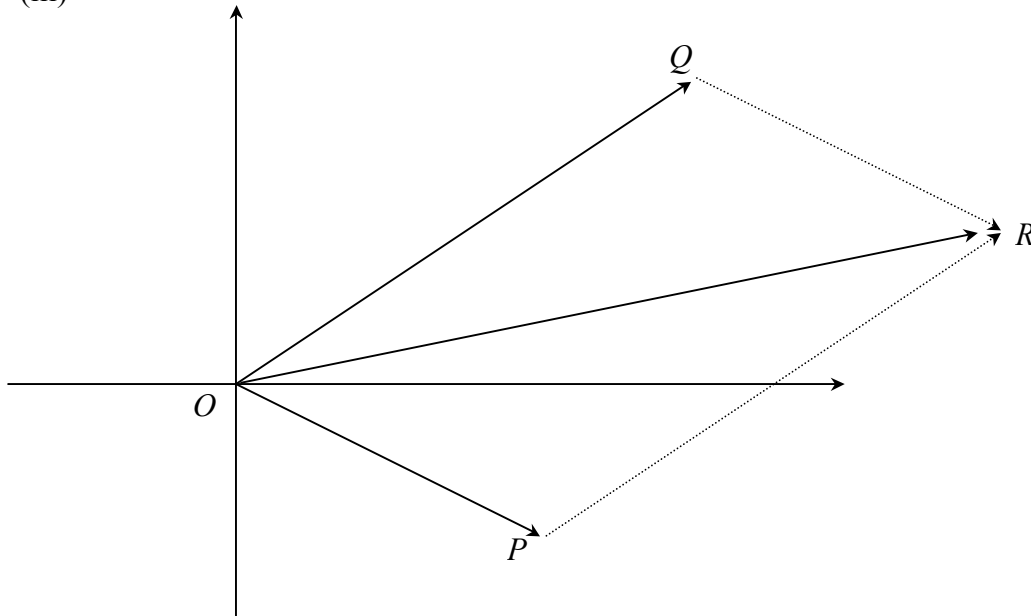
$$\overrightarrow{OS_1} = \overrightarrow{OQ} - \overrightarrow{S_1Q}$$

So S_1 is represented by $6 + 4i - [1 - 3\sqrt{3} + i(3 + \sqrt{3})] = 5 + 3\sqrt{3} + i(1 - \sqrt{3})$

$$\overrightarrow{OS_2} = \overrightarrow{OQ} - \overrightarrow{S_2Q}$$

So S_2 is represented by $6 + 4i - [1 + 3\sqrt{3} + i(3 - \sqrt{3})] = 5 - 3\sqrt{3} + i(1 + \sqrt{3})$

(iii)



$$\overrightarrow{OR} = \overrightarrow{OQ} + \overrightarrow{OP} = \overrightarrow{OQ} + \overrightarrow{OP}$$

So R corresponds to $6 + 4i + 4 - 2i = 10 + 2i$